

HR Metrics & Predictive Analytics

Employee Attrition Prediction Using Logistic Regression

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1. Business Question

Can employee-level HR metrics be used to predict whether an employee is likely to resign? This project treats voluntary resignation as the attrition outcome and demonstrates a complete predictive analytics workflow from preprocessing to model evaluation.

2. Dataset Overview

Item	Value
Original employee records	20,000
Active employees	17,360
Resigned employees	2,200
Terminated employees	440
Records used for attrition model	19,560
Attrition rate in modeling data	11.2%

3. Target and Predictor Variables

Target variable: Attrition = 1 for Resigned and 0 for Active. Terminated records were excluded because termination is a different employee-exit outcome. Predictors include department, performance rating, experience, work mode, salary, country, age, job level, tenure, performance score, salary band and hire year.

4. Data Preprocessing

- Filtered the data to Active and Resigned employees.
- Created a binary Attrition target.
- Separated numerical and categorical predictors.
- Standardized numerical variables.
- Applied one-hot encoding to categorical variables.
- Used a stratified 80/20 train-test split so the attrition proportion was preserved.
- Used class_weight='balanced' because resigned employees are the smaller class.

5. Model Selection

Logistic regression was selected because it is simple, interpretable and appropriate for a binary outcome. It provides predicted probabilities that can be evaluated with ROC-AUC and also produces coefficients that can help explain the direction of relationships. The model was evaluated using a held-out test set and 5-fold stratified cross-validation.

6. Model Evaluation

Metric	Test Result
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Accuracy	0.517
Precision	0.117
Recall	0.502
F1-score	0.190
ROC-AUC	0.496
5-fold CV mean ROC-AUC	0.491

The confusion matrix was TN=1,740, FP=1,760, FN=219, TP=220. The ROC-AUC of 0.496 is close to 0.50, and the 5-fold cross-validation mean is 0.491. This indicates that the available variables provide very limited predictive signal for resignation in this dataset.

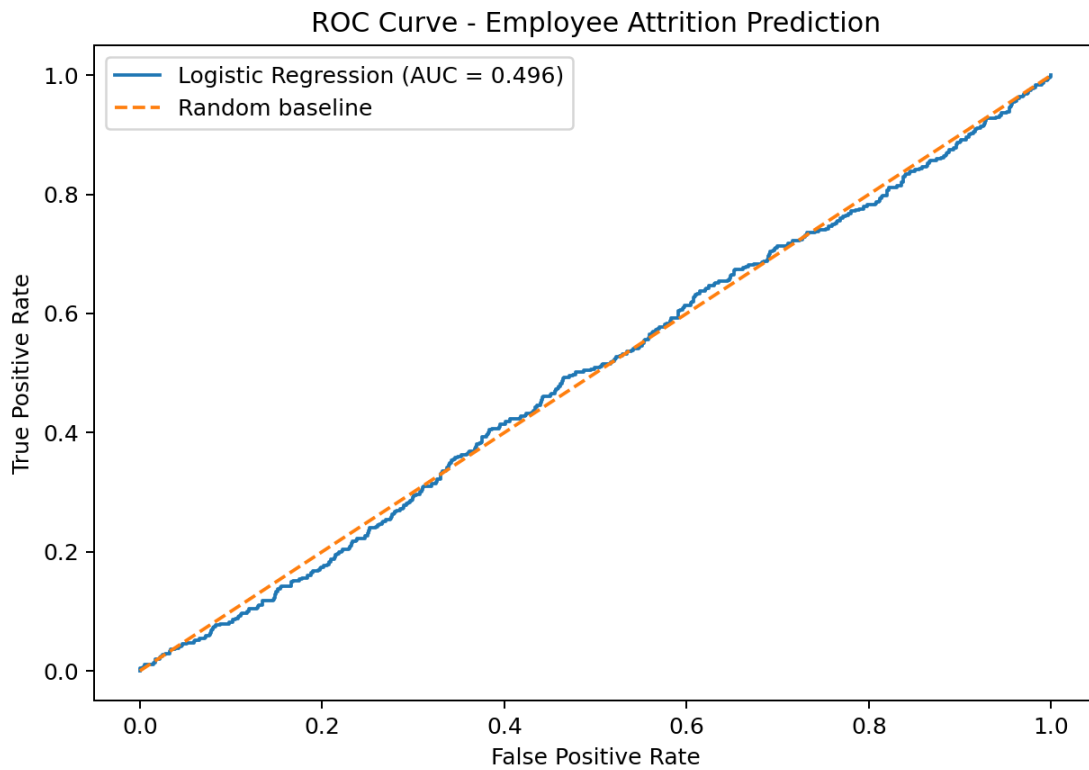


Figure 1. ROC curve for the attrition model.

7. Confusion Matrix

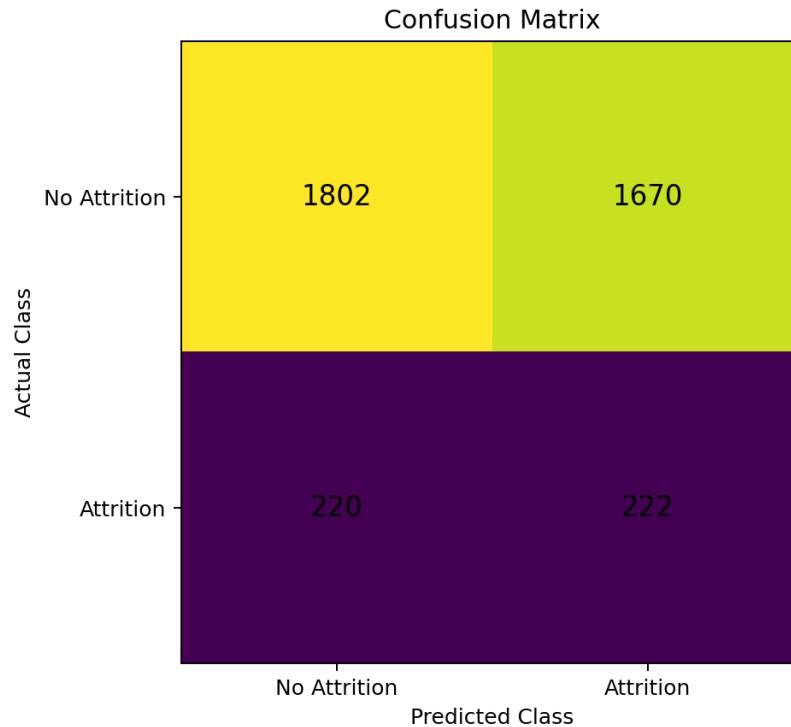


Figure 2. Test-set confusion matrix.

8. Interpretation

The model identifies about half of the resigned employees (recall ≈ 0.50), but its precision is low (≈ 0.12), meaning many employees predicted as likely to resign did not actually resign. More importantly, ROC-AUC is approximately 0.50, which is close to random classification. Therefore, this model should not be used as a production decision tool. Its main value in this project is demonstrating the complete predictive modeling workflow and showing that the current variables do not strongly explain attrition.

9. Relation to Previous HR Analysis

The earlier statistical analysis found that job level was strongly associated with salary, while experience and tenure had near-zero relationships with salary/performance and job level was not significantly associated with employment status. These findings help explain why a simple model based mainly on the existing HR fields may have limited attrition-prediction power.

10. HR Recommendations

- Do not use the current model to make hiring, firing, promotion or employee-treatment decisions.
- Collect stronger attrition-related variables such as engagement score, satisfaction, overtime, absenteeism, manager changes, promotion history and compensation changes if ethically and legally appropriate.
- Track resignation reasons and exit-interview themes because the current dataset does not directly capture why employees leave.
- Retrain and validate the model after additional predictive variables are collected.

- Use the existing Power BI dashboard for descriptive monitoring of workforce status, compensation, performance and hiring trends.

11. Conclusion

This project completed a practical HR predictive analytics pipeline: defining an attrition question, preparing the dataset, encoding and scaling variables, training logistic regression, validating the model, and evaluating it with accuracy, precision, recall, F1, ROC-AUC, cross-validation, a ROC curve and a confusion matrix. The final result is intentionally interpreted with caution: the model's near-random ROC-AUC shows that better HR predictors are required before predictive attrition analytics can be used reliably.